Mathematics of Data Science
Course Project — Paris Duchesse

Franck CHEN - Clémence JIN
M1 IDD — Université Paris Dauphine–PSL

Academic Year 2025–2026

Introduction

This document presents our approach to the Paris Duchesse office allocation problem. The objective is to minimize the total number of office moves throughout the renovation phases while satisfying operational and organizational constraints.

1 Linear Programming Relaxation

Question 1

Write a linear program corresponding to the problem. Indicate the number of variables and constraints, and write down the associated KKT conditions.

1.1 Decision Variables

Main variables.

$$z_{k,i,p} \in [0,1] \quad \forall k \in \{0,\dots,17\},\ i \in \{0,\dots,11\},\ p \in \{0,\dots,5\}$$

where $z_{k,i,p}$ indicates whether person k occupies office i at phase p.

Auxiliary variables for linearization.

$$d_{k,i,p} \geq 0 \quad \forall k \in \{0,\dots,17\},\ i \in \{0,\dots,11\},\ p \in \{0,\dots,4\}$$

where $d_{k,i,p}$ captures the move of person k from office i between phases p and $p+1$.

Total number of variables:

- Variables z: $18 \times 12 \times 6 = 1296$
- Variables d: $18 \times 12 \times 5 = 1080$
- **Total: 2376 variables**

1.2 Linearization of the Absolute Value

Instead of directly minimizing $|z_{k,i,p} - z_{k,i,p+1}|$ (which is non-linear), we introduce the variables $d_{k,i,p}$ with the constraints:

$$\begin{cases} d_{k,i,p} \geq z_{k,i,p} - z_{k,i,p+1}, \\ d_{k,i,p} \geq z_{k,i,p+1} - z_{k,i,p}. \end{cases}$$

These constraints ensure that

$$d_{k,i,p} \geq |z_{k,i,p} - z_{k,i,p+1}|.$$

Since the objective minimizes $d_{k,i,p}$, at optimality we have:

$$d_{k,i,p} = |z_{k,i,p} - z_{k,i,p+1}|.$$

1.3 Objective Function

$$\min \ \frac{1}{2} \sum_{k=0}^{17} \sum_{i=0}^{11} \sum_{p=0}^{4} d_{k,i,p}$$

Justification of the factor $\frac{1}{2}$: A move of person k from office i to office j between phases p and $p+1$ is counted twice:

- once in $d_{k,i,p}$ (departure from office i),
- once in $d_{k,j,p}$ (arrival to office j).

The factor $\frac{1}{2}$ corrects this double counting.

1.4 Constraints

C1. Unique Location Each person must occupy exactly one office at each phase:

$$\sum_{i=0}^{11} z_{k,i,p} = 1 \quad \forall k \in \{0, \ldots, 17\}, \ \forall p \in \{0, \ldots, 5\}.$$

Count: $18 \times 6 = 108$ constraints.

C2. Office Capacity Each office can accommodate at most two people:

$$\sum_{k=0}^{17} z_{k,i,p} \leq 2 \quad \forall i \in \{0, \ldots, 11\}, \ \forall p \in \{0, \ldots, 5\}.$$

Count: $12 \times 6 = 72$ constraints.

C3. Linearization Constraints To capture moves:

$$\begin{cases} d_{k,i,p} \geq z_{k,i,p} - z_{k,i,p+1}, \\ d_{k,i,p} \geq z_{k,i,p+1} - z_{k,i,p}, \end{cases} \quad \forall k \in \{0, \ldots, 17\}, \ i \in \{0, \ldots, 11\}, \ p \in \{0, \ldots, 4\}.$$

Count: $2 \times 18 \times 12 \times 5 = 2160$ constraints.

C4. Closed Offices Some offices are closed at certain phases (renovation or not yet built):

$$z_{k,i,p} = 0 \quad \forall k \in \{0, \ldots, 17\}, \quad (i,p) \in \mathcal{F}.$$

where $\mathcal{F}$ is the set of closed (office, phase) pairs:

$$\mathcal{F} = \{(E1,0), (E2,0), (B1,1), (B2,1), (B3,1), (D1,2), (D2,2), (D3,2), (C1,3), (C2,3), (A1,4), (A2,4)\}.$$

Count: $18 \times 12 = 216$ constraints.

C5. Initial Positions (Phase 0) For each person k, the initial office b_k is fixed:

$$z_{k,b_k,0} = 1 \quad \forall k \in \{0, \ldots, 17\}.$$

Count: 18 constraints.

C6. Final Assignments by Service (Phase 5) Each person must be assigned to exactly one office within their allocated service:

$$\sum_{i \in \text{Offices}(\text{Service}(k))} z_{k,i,5} = 1 \quad \forall k \in \{0, \ldots, 17\}$$

where $\text{Offices}(\text{Service}(k))$ denotes the set of offices assigned to the service of person k.

Additional Distribution Constraints:

Mathematics Service (M), Wing B:

$$\sum_{k \in M} z_{k,B1,5} = 1, \qquad \sum_{k \in M} z_{k,B2,5} = 2$$

Theoretical CS Service (T), Wing D:

$$\sum_{k \in T} z_{k,D1,5} = 1, \qquad \sum_{k \in T} z_{k,D2,5} = 2$$

Optimization Service (O), Wing E:

$$\sum_{k \in O} z_{k,E1,5} = 1$$

Note: The constraints on offices B3, D3 and E2 are implied by the previous constraints.

Count: $18 + 5 = 23$ constraints.

C7. Bound Constraints

$$0 \leq z_{k,i,p} \leq 1, \quad d_{k,i,p} \geq 0.$$

Total bound constraints: 3672.

Implementation Remark: In CVXPY, non-negativity constraints are handled using `nonneg=True`. The constraint $z \leq 1$ is added explicitly and represents 1296 scalar constraints.

1.5 Summary Table

Element	Count
Variables z	1296
Variables d	1080
Total variables	**2376**
Constraints C1–C6	2597
$z \leq 1$ (C7 partial)	1296
Subtotal	**3893**
$z \geq 0$, $d \geq 0$ (C7 partial)	2376
Total constraints	**6269**

1.6 KKT Conditions

1.6.1 Standard Formulation of the Primal Problem

The primal problem can be written in standard form:

$$\begin{aligned} \min_{z,d} \quad & f(z,d) = \frac{1}{2} \sum_{k,i,p} d_{k,i,p} \\ \text{s.t.} \quad & h_j(z,d) = 0, \quad j = 1, \ldots, m_{eq} \quad \text{(equality constraints)} \\ & g_\ell(z,d) \leq 0, \quad \ell = 1, \ldots, m_{ineq} \quad \text{(inequality constraints)} \end{aligned}$$

where $x = (z,d) \in \mathbb{R}^{2376}$ is the primal variable vector.

Equality constraints h_j (total: $m_{eq} = 365$ constraints):

- **C1** (location uniqueness): 108 constraints

$$h^{C1}_{k,p}(z) = \sum_{i=0}^{11} z_{k,i,p} - 1 = 0 \quad \forall k \in \{0, \ldots, 17\},\ p \in \{0, \ldots, 5\}$$

- **C4** (closed offices): 216 constraints

$$h^{C4}_{k,i,p}(z) = z_{k,i,p} = 0 \quad \forall k \in \{0, \ldots, 17\},\ \forall (i,p) \in \mathcal{F}$$

  where

$$\begin{aligned} \mathcal{F} = \{&(E1,0), (E2,0), (B1,1), (B2,1), (B3,1), (D1,2), \\ &(D2,2), (D3,2), (C1,3), (C2,3), (A1,4), (A2,4)\} \end{aligned}$$

- **C5** (initial positions): 18 constraints

$$h^{C5}_k(z) = z_{k,b_k,0} - 1 = 0 \quad \forall k \in \{0, \ldots, 17\}$$

  where b_k is the initial office of person k

- **C6** (final assignments): 23 constraints

$$h^{C6}_k(z) = \sum_{i \in \text{Offices}(\text{Service}(k))} z_{k,i,5} - 1 = 0 \quad \forall k \in \{0, \ldots, 17\} \quad \text{(18 constraints)}$$

$$h^{C6}_{\text{dist}}(z) = \sum_{k \in \text{Service}} z_{k,i,5} - n_i = 0 \quad \text{for certain offices } i \quad \text{(5 constraints)}$$

  where $n_i \in \{1, 2\}$ is the exact number of people required in office i and "dist" stands for distribution

Inequality constraints g_ℓ (total: $m_{ineq} = 5904$ constraints):

- **C2** (office capacity): 72 constraints

$$g_{i,p}^{C2}(z) = \sum_{k=0}^{17} z_{k,i,p} - 2 \leq 0 \quad \forall i \in \{0, \ldots, 11\},\ p \in \{0, \ldots, 5\}$$

- **C3** (linearization): 2160 constraints

$$g_{k,i,p}^{C3,+}(z, d) = z_{k,i,p} - z_{k,i,p+1} - d_{k,i,p} \leq 0$$
$$g_{k,i,p}^{C3,-}(z, d) = z_{k,i,p+1} - z_{k,i,p} - d_{k,i,p} \leq 0$$

  $\forall k \in \{0, \ldots, 17\},\ i \in \{0, \ldots, 11\},\ p \in \{0, \ldots, 4\}$

- **C7** (bound constraints): 3672 constraints

$$g_{k,i,p}^{z_L}(z) = -z_{k,i,p} \leq 0 \qquad \text{(lower bounds on } z\text{, 1296 constraints)}$$
$$g_{k,i,p}^{z_U}(z) = z_{k,i,p} - 1 \leq 0 \qquad \text{(upper bounds on } z\text{, 1296 constraints)}$$
$$g_{k,i,p}^{d_L}(d) = -d_{k,i,p} \leq 0 \qquad \text{(lower bounds on } d\text{, 1080 constraints)}$$

1.6.2 Lagrangian

The Lagrangian of the problem is:

$$L(z, d, \lambda, \mu) = \frac{1}{2} \sum_{k,i,p} d_{k,i,p} + \sum_{j=1}^{365} \lambda_j h_j(z, d) + \sum_{\ell=1}^{5904} \mu_\ell g_\ell(z, d)$$

where:

- $\lambda = (\lambda_1, \ldots, \lambda_{365}) \in \mathbb{R}^{365}$: multipliers of equality constraints (free)
- $\mu = (\mu_1, \ldots, \mu_{5904}) \in \mathbb{R}_+^{5904}$: multipliers of inequality constraints (non-negative)

1.6.3 Karush-Kuhn-Tucker Theorem

Theorem: Let (z^*, d^*) be a local solution of the primal problem and suppose that the active constraints at (z^*, d^*) satisfy the **Mangasarian-Fromovitz constraint qualification** (MFCQ). Then there exist Lagrange multipliers (λ^*, μ^*) such that the following KKT conditions are satisfied:

1. Stationarity (zero gradients):

$$\nabla_x L(z^*, d^*, \lambda^*, \mu^*) = \nabla f(z^*, d^*) + \sum_{j=1}^{365} \lambda_j^* \nabla h_j(z^*, d^*) + \sum_{\ell=1}^{5904} \mu_\ell^* \nabla g_\ell(z^*, d^*) = 0$$

2. Primal feasibility:

$$h_j(z^*, d^*) = 0, \qquad \forall j = 1, \ldots, 365$$
$$g_\ell(z^*, d^*) \leq 0, \qquad \forall \ell = 1, \ldots, 5904$$

3. Dual feasibility:

$$\lambda_j^* \in \mathbb{R}, \qquad \forall j = 1, \ldots, 365$$
$$\mu_\ell^* \geq 0, \qquad \forall \ell = 1, \ldots, 5904$$

4. Complementarity:

$$\mu_\ell^* \cdot g_\ell(z^*, d^*) = 0, \quad \forall \ell = 1, \ldots, 5904$$

1.6.4 Applicability to the Relocation Problem

Verification of constraint qualification:
For our linear problem:

- $f(z, d) = \frac{1}{2} \sum d_{k,i,p}$ is **linear**
- All constraints h_j and g_ℓ are **affine** (linear)

Linear constraints automatically satisfy the MFCQ condition (in fact, all standard constraint qualifications). Moreover:

Proposition: For a linear problem, the KKT conditions are **necessary and sufficient** for global optimality.
Thus, solving our problem is equivalent to finding $(z^*, d^*, \lambda^*, \mu^*)$ satisfying the 4 KKT conditions.

1.6.5 Explicit KKT Conditions for Our Problem

Stationarity for $z_{k,i,p}$:

$$\frac{\partial L}{\partial z_{k,i,p}} = \lambda^{C1}_{k,p} + \mathbb{1}_{(i,p)\in\mathcal{F}}\lambda^{C4}_{k,i,p} + \mathbb{1}_{p=0,i=b_k}\lambda^{C5}_{k} + \mathbb{1}_{p=5}\lambda^{C6}_{k} + \mu^{C2}_{i,p} + \Delta^{C3}_{k,i,p} - \mu^{z_L}_{k,i,p} + \mu^{z_U}_{k,i,p} = 0$$

where:

$$\Delta^{C3}_{k,i,p} = \begin{cases} \mu^{C3,+}_{k,i,p} - \mu^{C3,-}_{k,i,p} - \mu^{C3,+}_{k,i,p-1} + \mu^{C3,-}_{k,i,p-1} & \text{if } 1 \le p \le 4 \\ \mu^{C3,+}_{k,i,0} - \mu^{C3,-}_{k,i,0} & \text{if } p = 0 \\ -\mu^{C3,+}_{k,i,4} + \mu^{C3,-}_{k,i,4} & \text{if } p = 5 \end{cases}$$

represents the contribution of linearization constraints C3.

Stationarity for $d_{k,i,p}$:

$$\frac{\partial L}{\partial d_{k,i,p}} = \frac{1}{2} - \mu^{C3,+}_{k,i,p} - \mu^{C3,-}_{k,i,p} - \mu^{d_L}_{k,i,p} = 0$$

Complementarity (key examples):

$$\begin{aligned}
\mu^{C2}_{i,p} \cdot \left(\sum_{k=0}^{17} z^*_{k,i,p} - 2 \right) &= 0 && \forall i, p \\
\mu^{C3,+}_{k,i,p} \cdot (z^*_{k,i,p} - z^*_{k,i,p+1} - d^*_{k,i,p}) &= 0 && \forall k, i, p \in \{0, \dots, 4\} \\
\mu^{C3,-}_{k,i,p} \cdot (z^*_{k,i,p+1} - z^*_{k,i,p} - d^*_{k,i,p}) &= 0 && \forall k, i, p \in \{0, \dots, 4\} \\
\mu^{z_L}_{k,i,p} \cdot z^*_{k,i,p} &= 0 && \forall k, i, p \\
\mu^{z_U}_{k,i,p} \cdot (z^*_{k,i,p} - 1) &= 0 && \forall k, i, p \\
\mu^{d_L}_{k,i,p} \cdot d^*_{k,i,p} &= 0 && \forall k, i, p \in \{0, \dots, 4\}
\end{aligned}$$

1.6.6 Summary Table of Dual Multipliers

Verification: The total number of multipliers (6269) equals the total number of constraints, confirming that each constraint of the primal problem has an associated dual multiplier.

Multiplier type	Associated with	Dimension	Sign
$\lambda^{C1}_{k,p}$	Equality: uniqueness	$18 \times 6 = 108$	Free
$\lambda^{C4}_{k,i,p}$	Equality: closed offices	216	Free
λ^{C5}_{k}	Equality: initial positions	18	Free
λ^{C6}	Equality: final assignments	23	Free
Subtotal λ	**Equality constraints**	**365**	**Free**
$\mu^{C2}_{i,p}$	Inequality: capacity	$12 \times 6 = 72$	≥ 0
$\mu^{C3,+}_{k,i,p}$	Inequality: linearization (+)	$18 \times 12 \times 5 = 1080$	≥ 0
$\mu^{C3,-}_{k,i,p}$	Inequality: linearization (-)	$18 \times 12 \times 5 = 1080$	≥ 0
$\mu^{z_L}_{k,i,p}$	Inequality: $z \geq 0$	$18 \times 12 \times 6 = 1296$	≥ 0
$\mu^{z_U}_{k,i,p}$	Inequality: $z \leq 1$	$18 \times 12 \times 6 = 1296$	≥ 0
$\mu^{d_L}_{k,i,p}$	Inequality: $d \geq 0$	$18 \times 12 \times 5 = 1080$	≥ 0
Subtotal μ	**Inequality constraints**	**5904**	≥ 0
Total multipliers		**6269**	

Question 2

Compute a solution for your problem and its dual.

2.1 Solution of the Linear Relaxation

The primal problem was solved using the ECOS solver in CVXPY. The results are:

Metric	Value
Solution status	Optimal
Objective value	30.0 moves
Solver	ECOS

Interpretation:

- The optimal value of 30.0 moves represents a **lower bound** for the integer problem.
- Since this is a linear relaxation, the solution contains **fractional assignments**, where some $z_{k,i,p}$ values lie strictly between 0 and 1.
- This means some people are "partially" assigned to multiple offices at certain phases, which is not physically realizable.
- Any valid integer solution must require at least 30 moves.

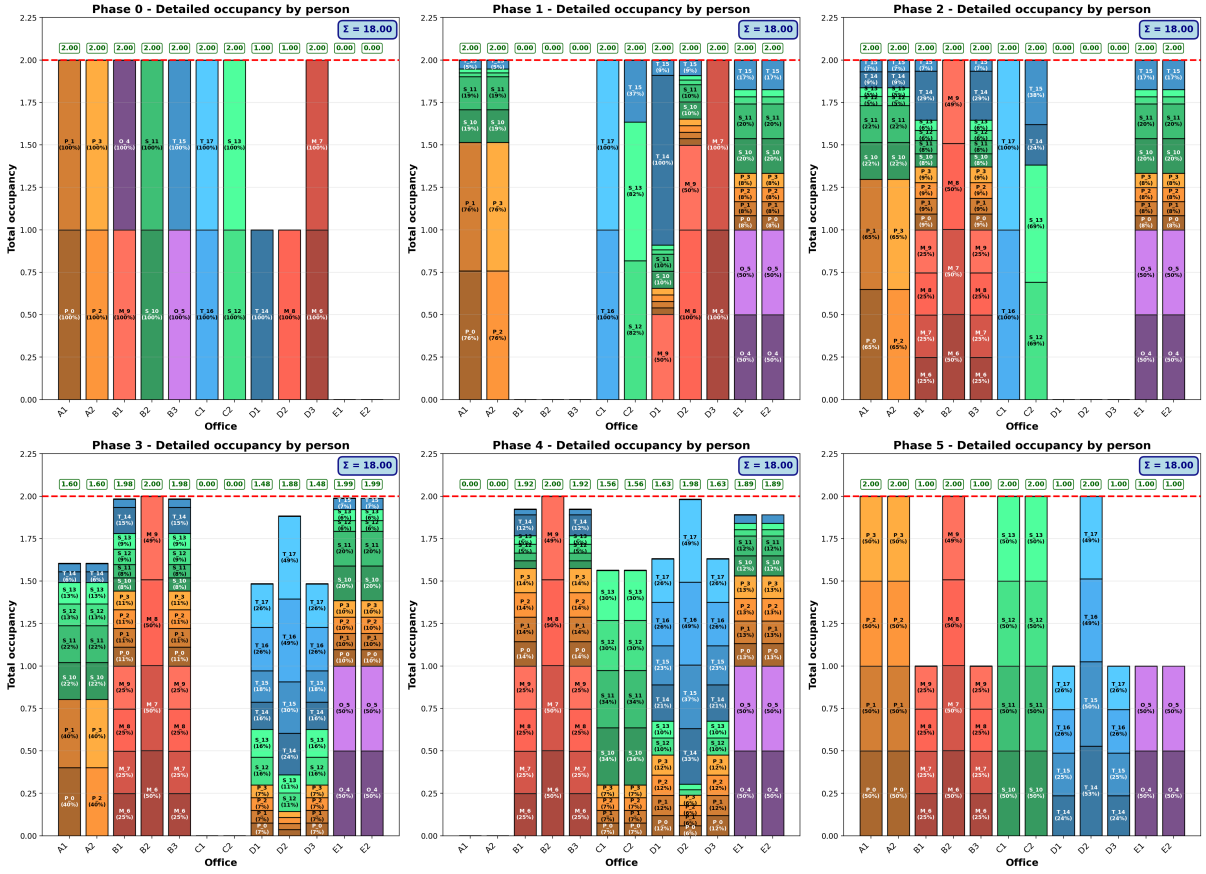


Figure 1: Office occupancy by phase for LP

Histogram Analysis:

Figure 1 shows the detailed office occupancy by phase. For each office (A1, A2, B1, etc.) and each phase, the stacked bar chart displays:

- **Total occupancy:** The height of each bar represents the total number of people assigned to that office (ranging from 0 to 2.25), displayed numerically above each bar
- **Individual person contributions:** Each bar is decomposed by stacked colored segments, where each segment represents one person's fractional assignment. For example, a segment labeled "M_6 (50%)" indicates that person 6 from the Mathematics service is assigned with weight 0.5 to that office
- **Color coding by service:** Each service has a base color (Mathematics: red, Presidency: orange, Theoretical CS: blue, Students: green, Optimization: purple), with shades distinguishing individual people within each service
- **Fractional assignments:** Percentages less than 100% reveal the fractional splitting in the linear relaxation, where people are partially assigned to multiple offices simultaneously
- **Capacity violations:** Offices exceeding the capacity of 2 people ($> 100\% + 100\%$) are highlighted with a red dotted border and a warning symbol, with their total occupancy shown in red
- **Total sum per phase:** The top-right corner of each subplot displays Σ, the total number of people across all offices, which should equal 18

This visualization clearly demonstrates why integrality constraints are necessary to obtain a practical relocation plan with whole-person assignments.

2.2 Dual Solution and Strong Duality Verification

The dual problem was solved simultaneously with the primal problem. The dual variables (Lagrange multipliers) provide important economic interpretations of the constraints.

Constraint	**RHS** (b)	**Dual Contribution**
C1 (Uniqueness)	1	13.82
C2 (Capacity)	2	−19.54
C3 (Linearization)	0	0.00
C4 (Closed offices)	0	0.00
C5 (Initial positions)	1	22.56
C6 (Final assignments)	variable	17.59
C7 (Upper bounds $z \leq 1$)	1	−4.42
Total dual objective		**30.00**

Strong Duality Verification:

- Primal objective value: 30.0000000000
- Dual objective value: 29.9999999995
- Gap: 5.27×10^{-10} (negligible numerical error)
- **Strong duality verified**

Economic Interpretation:
The dual contributions reveal the "cost" or "value" of each constraint type:

- **C5 (Initial positions, +22.56):** The largest positive contribution indicates that fixing initial positions significantly increases the objective. Relaxing these constraints would allow reducing the number of moves.
- **C2 (Capacity, −19.54):** The large negative contribution shows that capacity constraints are binding and valuable. Increasing office capacity from 2 to 3 would reduce the optimal objective by approximately 19.54 moves.
- **C6 (Final assignments, +17.59):** The positive contribution indicates that the final assignment requirements add significant constraints to the problem.
- **C1 (Uniqueness, +13.82):** Enforcing that each person must be in exactly one office per phase adds cost to the solution.
- **C7 (Upper bounds, −4.42):** The negative contribution suggests that the upper bound constraints $z_{k,i,p} \leq 1$ are active and reducing them would decrease the objective.
- **C3, C4 (Zero contribution):** These constraints are satisfied with strict inequality in the optimal solution (non-binding constraints).

Complementary Slackness:
Since strong duality holds with zero gap, the complementary slackness conditions are satisfied:

- For active constraints (with non-zero dual values), the primal constraint is satisfied with equality
- For inactive constraints (with zero dual values), the primal constraint has slack

This dual analysis provides valuable insights for potential problem modifications, such as increasing office capacity or relaxing certain assignment constraints.

Question 3

Modify the problem to avoid adjacency between the students' association and the presidency. Compute a solution and compare it with Question 2.

3.1 Definition of Adjacent Offices

Two offices are considered adjacent if they share a common wall or are directly connected. In the Paris Duchesse University building, the pairs of adjacent offices include:

Adjacencies within wings:

- Wing A: $(A1, A2)$
- Wing B: $(B1, B2)$ and $(B2, B3)$
- Wing C: $(C1, C2)$
- Wing D: $(D1, D2)$ and $(D2, D3)$
- Wing E: $(E1, E2)$

Adjacencies between wings:

- $(D2, E1)$, $(E2, B2)$
- $(D1, C2)$, $(C1, B3)$, $(B1, A2)$, $(D3, A1)$

Let $\mathcal{A}$ be the set of adjacent office pairs:

$$\begin{aligned}\mathcal{A} = \{&(A1, A2), (B1, B2), (B2, B3), (C1, C2), (D1, D2), (D2, D3), (E1, E2),\\ &(D2, E1), (E2, B2), (D1, C2), (C1, B3), (B1, A2), (D3, A1)\}\end{aligned}$$

We have $|\mathcal{A}| = 13$ adjacent pairs.

3.2 Constraint Formulation with Binary Variables

To model the mutual exclusion of P and S in adjacent offices, we use a **Big-M** formulation with binary variables.

Decision variables:

For each pair of adjacent offices $(i, j) \in \mathcal{A}$ and each phase $p \in \{0, 1, 2, 3, 4, 5\}$, we introduce a binary variable:

$$y_{(i,j),p} \in \{0, 1\}$$

where:

- $y_{(i,j),p} = 1$: the Presidency service can occupy the pair (i, j) at phase p (Students cannot)

- $y_{(i,j),p} = 0$: the Students service can occupy the pair (i, j) at phase p (Presidency cannot)

Big-M constraints:

For each pair $(i, j) \in \mathcal{A}$ and each phase p:

$$\sum_{k \in P} z_{k,i,p} + \sum_{k \in P} z_{k,j,p} \leq M \cdot y_{(i,j),p}$$
$$\sum_{k \in S} z_{k,i,p} + \sum_{k \in S} z_{k,j,p} \leq M \cdot (1 - y_{(i,j),p})$$

where M is a sufficiently large constant (here $M = 4$ since each office can contain at most 2 people).

Interpretation:

- If $y_{(i,j),p} = 1$: the first constraint is relaxed (P can occupy the offices), while the second forces $\sum_{k \in S}(z_{k,i,p} + z_{k,j,p}) = 0$ (no member of S)
- If $y_{(i,j),p} = 0$: conversely, S can occupy the pair but not P

This ensures that at any time, for each pair of adjacent offices, only one of the two services (P or S) can be present.

3.3 Problem Complexity

1) Number of additional binary variables:

$$|\mathcal{A}| \times \text{number of phases} = 13 \times 6 = 78 \text{ binary variables}$$

2) Number of additional constraints:

$$2 \times |\mathcal{A}| \times \text{number of phases} = 2 \times 13 \times 6 = 156 \text{ constraints}$$

Problem type:

- The problem becomes a **Mixed-Integer Linear Program (MILP)**
- The variables $z_{k,i,p}$ remain continuous in $[0, 1]$ (linear relaxation)
- The variables $y_{(i,j),p}$ are strictly binary $\{0, 1\}$

3.4 Implementation and Results

The MILP problem was solved using the CBC (COIN-OR Branch and Cut) solver through CVXPY. The computational results are presented below.

Metric	Without Separation (LP)	With Separation (MILP)
Solver	ECOS	CBC
Problem type	Linear Program	Mixed-Integer LP
Status	Optimal	Optimal
Objective value	30.000000	30.000000
Number of moves	30.00	30.00

Impact of the separation constraint:

- Objective value increase: 0.00%
- Additional moves required: 0.00
- The separation constraint has **no impact** on the optimal objective value

3.5 Surprising Discovery: Integer Solution from Linear Relaxation

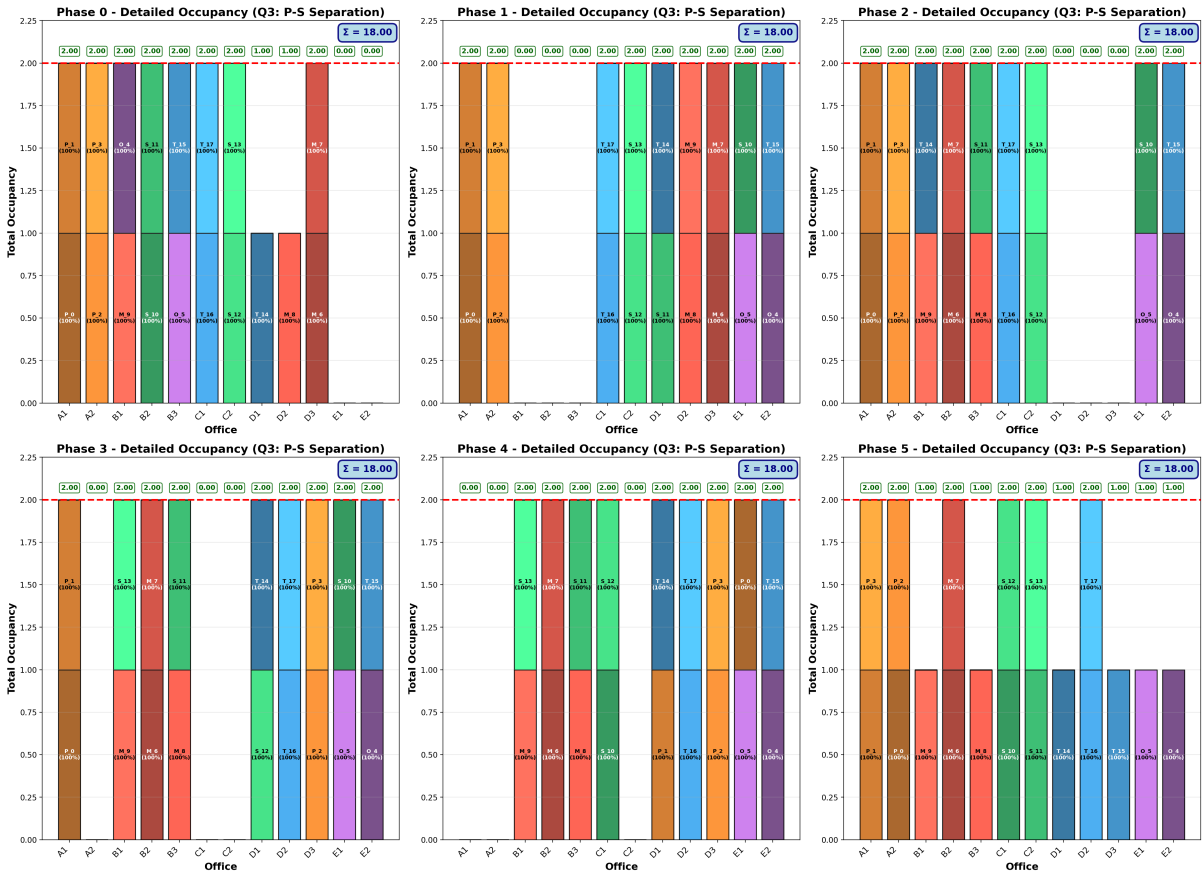


Figure 2: Office occupancy by phase for MILP

The most remarkable observation is that the MILP solution is **entirely integer**, despite the fact that the $z_{k,i,p}$ variables were only constrained to $[0, 1]$ (not forced to be binary).

Figure 2 shows the office occupancy visualization for the MILP solution. Unlike the pure LP relaxation (Figure 1), this histogram reveals:

- **All assignments are integer:** Every $z_{k,i,p} \in \{0, 1\}$, with no fractional values
- **Whole-person assignments:** Each person is assigned to exactly one office per phase (no splitting)
- **Integer occupancy levels:** All offices contain 0, 1, or 2 people (never fractional values like 1.5)
- **Practical feasibility:** This solution is directly implementable without any rounding or post-processing

Theoretical Explanation:

This phenomenon occurs because:

1. **Total unimodularity:** The constraint matrix of the relocation problem (excluding the Big-M constraints) has special structure that often leads to integer solutions even in the LP relaxation

2. **Binary constraint propagation:** The introduction of binary variables $y_{(i,j),p} \in \{0, 1\}$ creates a **forcing effect**. When $y = 1$ or $y = 0$, the Big-M constraints become:

   - Either $\sum_{k \in P}(z_{k,i,p} + z_{k,j,p}) \leq 4$ and $\sum_{k \in S}(z_{k,i,p} + z_{k,j,p}) = 0$
   - Or $\sum_{k \in P}(z_{k,i,p} + z_{k,j,p}) = 0$ and $\sum_{k \in S}(z_{k,i,p} + z_{k,j,p}) \leq 4$

   These zero-forcing constraints, combined with the uniqueness constraints (C1), push the continuous variables z toward binary values.

3. **Problem geometry:** The feasible region of this particular MILP has vertices that coincide with integer points, so the linear objective function naturally attains its optimum at an integer vertex

4. **Integrality gap = 0:** The LP relaxation bound (30.0) equals the integer optimum (30.0), indicating that the relaxation is **exact** for this instance

Practical Implications:

- The separation constraint can be enforced **at no additional cost** in terms of the number of moves
- The solution is **immediately implementable** without rounding
- This suggests the original problem structure is "well-behaved" and naturally admits integer solutions
- For this specific instance, solving the MILP is equivalent to solving a full Integer Linear Program (ILP), even though we only enforced integrality on the y variables

Important caveat: This integer property is not guaranteed for all instances. Different initial configurations, capacities, or separation requirements might yield fractional LP relaxations, requiring explicit integrality constraints on $z_{k,i,p}$ or branch-and-bound methods to obtain integer solutions.

The non-adjacency constraint between the Presidency and Students association was not maintained in Parts 2 and 3 to simplify the formulation and focus on the core optimization techniques (penalization and semidefinite relaxation), as this constraint significantly complicates the problem structure without adding substantial pedagogical value to these specific sections.

2 Penalized Problem

Question 4

Adapt the linear program to include the penalty term and give the dual formulation.

4.1 Motivation

In Question 1, the objective was solely to minimize the total number of relocations. However, it may be desirable to reach the final allocation **as early as possible** during the renovation process.

4.2 Penalty Formulation

To encourage rapid convergence toward the final allocation z^F (Phase 5), we add a penalty term to the objective function:

$$\lambda \sum_{p=0}^{5} \|z^p - z^F\|_1 = \lambda \sum_{p=0}^{5} \sum_{k=0}^{17} \sum_{i=0}^{11} |z_{k,i,p} - z_{k,i,5}|$$

where:

- $z^p = (z_{k,i,p})_{k,i}$: allocation vector at phase p
- $z^F = z^5 = (z_{k,i,5})_{k,i}$: final allocation vector (phase 5)
- $\lambda > 0$: **penalty weight** controlling the relative importance of this constraint
- $\|\cdot\|_1$: L^1 norm (sum of absolute values)

Interpretation:

- At each phase p, we measure the "distance" between the current allocation and the final allocation
- The sum over all phases encourages reducing this distance as soon as possible
- A high λ prioritizes rapid convergence at the potential expense of the number of relocations

4.3 New Objective Function

The penalized objective function becomes:

$$\min_{z,d,e} \quad \frac{1}{2} \sum_{k,i,p} d_{k,i,p} + \lambda \sum_{p,k,i} e_{k,i,p}$$

where $e_{k,i,p} \geq 0$ are **new auxiliary variables** to linearize the L^1 norm.

4.4 Linearization of the Penalty

For each k, i, p, we introduce $e_{k,i,p}$ with the constraints:

$$\begin{cases} e_{k,i,p} \geq z_{k,i,p} - z_{k,i,5} \\ e_{k,i,p} \geq z_{k,i,5} - z_{k,i,p} \end{cases}$$

These constraints guarantee $e_{k,i,p} \geq |z_{k,i,p} - z_{k,i,5}|$, and minimization ensures equality at optimum.

Number of new variables:

$$18 \times 12 \times 6 = 1296 \text{ variables } e$$

Number of new constraints:

$$2 \times 18 \times 12 \times 6 = 2592 \text{ linearization constraints}$$

4.5 Complete Problem with Penalty

The penalized linear program is written as:

$$
\begin{aligned}
\min_{z,d,e} \quad & \frac{1}{2}\sum_{k,i,p} d_{k,i,p} + \lambda \sum_{k,i,p} e_{k,i,p} \\
\text{s.t.} \quad & \text{Constraints C1–C7 (identical to Q1)} \\
& e_{k,i,p} \geq z_{k,i,p} - z_{k,i,5} && \forall k, i, p \\
& e_{k,i,p} \geq z_{k,i,5} - z_{k,i,p} && \forall k, i, p \\
& e_{k,i,p} \geq 0 && \forall k, i, p
\end{aligned}
$$

4.6 Trade-off: Relocations vs Convergence

The parameter λ controls the balance between two competing objectives:

$\lambda \to 0$ **(low penalty):**

- Priority to minimal number of relocations
- Convergence to z^F may be delayed
- Solution close to that of Q1

λ **moderate** ($\lambda = 100$ **in our study):**

- Balanced compromise
- Number of relocations slightly increased
- Faster convergence toward final allocation

$\lambda \to \infty$ **(high penalty):**

- Absolute priority to rapid convergence
- Number of relocations potentially very high
- Final allocation reached as soon as physically possible

4.7 Derivation of the Dual of the Penalized Problem

The penalized primal problem is written in standard form:

$$
\begin{aligned}
\min_{z,d,e} \quad & \frac{1}{2}\mathbf{1}^T d + \lambda \mathbf{1}^T e \\
\text{s.t.} \quad & A_{eq} \begin{pmatrix} z \\ d \\ e \end{pmatrix} = b_{eq} && \text{(equality constraints C1, C4, C5, C6)} \\
& A_{ineq} \begin{pmatrix} z \\ d \\ e \end{pmatrix} \leq b_{ineq} && \text{(inequality constraints C2, C3, C7, C8)} \\
& z, d, e \geq 0
\end{aligned}
$$

where A_{eq} and A_{ineq} now include the constraints related to e.

4.8 Structure of the Penalized Lagrangian

The Lagrangian of the penalized problem is:

$$L(z,d,e,\lambda,\mu) = \frac{1}{2}\sum_{k,i,p} d_{k,i,p} + \lambda \sum_{k,i,p} e_{k,i,p} + \sum_{j=1}^{365} \lambda_j h_j(z,d,e) + \sum_{\ell=1}^{m_{ineq}} \mu_\ell g_\ell(z,d,e)$$

where:

- The equality constraints h_j are identical to Q1/Q2 (C1, C4, C5, C6)
- The inequality constraints g_ℓ include:
  - C2 (capacity): 72 constraints
  - C3 (linearization of d): 2160 constraints
  - C7 (bounds on z, d): 3672 constraints
  - **C8 (linearization of e)**: 2592 new constraints

4.9 New Constraints C8 for the Penalty

For each $k \in \{0,\dots,17\}$, $i \in \{0,\dots,11\}$, $p \in \{0,\dots,5\}$:

$$g^{C8,+}_{k,i,p}(z,e) = z_{k,i,p} - z_{k,i,5} - e_{k,i,p} \le 0 \qquad \text{(1296 constraints)}$$
$$g^{C8,-}_{k,i,p}(z,e) = z_{k,i,5} - z_{k,i,p} - e_{k,i,p} \le 0 \qquad \text{(1296 constraints)}$$

These constraints guarantee $e_{k,i,p} \ge |z_{k,i,p} - z_{k,i,5}|$.

Associated dual multipliers:

- $\mu^{C8,+}_{k,i,p} \ge 0$: multiplier for $g^{C8,+}_{k,i,p}$
- $\mu^{C8,-}_{k,i,p} \ge 0$: multiplier for $g^{C8,-}_{k,i,p}$

4.10 Stationarity Conditions of the Lagrangian

Stationarity for $z_{k,i,p}$ (allocation primal variables):

$$\frac{\partial L}{\partial z_{k,i,p}} = \lambda^{C1}_{k,p} + \mathbb{1}_{(i,p)\in\mathcal{F}}\lambda^{C4}_{k,i,p} + \mathbb{1}_{p=0,i=b_k}\lambda^{C5}_{k} + \mathbb{1}_{p=5}\lambda^{C6}_{k} + \mu^{C2}_{i,p} + \Delta^{C3}_{k,i,p} + \Delta^{C8}_{k,i,p} - \mu^{z_L}_{k,i,p} + \mu^{z_U}_{k,i,p} = 0$$

where $\Delta^{C8}_{k,i,p}$ is the **new contribution** from the convergence penalty:

$$\Delta^{C8}_{k,i,p} = \begin{cases} \mu^{C8,+}_{k,i,p} - \mu^{C8,-}_{k,i,p} & \text{if } p \neq 5 \\ -\sum_{p'=0}^{4}(\mu^{C8,+}_{k,i,p'} - \mu^{C8,-}_{k,i,p'}) & \text{if } p = 5 \text{ (final allocation)} \end{cases}$$

Stationarity for $d_{k,i,p}$ (relocation variables):

Identical to Q2:

$$\frac{\partial L}{\partial d_{k,i,p}} = \frac{1}{2} - \mu^{C3,+}_{k,i,p} - \mu^{C3,-}_{k,i,p} - \mu^{d_L}_{k,i,p} = 0$$

Stationarity for $e_{k,i,p}$ (new penalty variables):

$$\frac{\partial L}{\partial e_{k,i,p}} = \lambda - \mu^{C8,+}_{k,i,p} - \mu^{C8,-}_{k,i,p} - \mu^{e_L}_{k,i,p} = 0$$

where $\mu^{e_L}_{k,i,p} \geq 0$ is the multiplier for the constraint $e_{k,i,p} \geq 0$.

4.11 Explicit Formulation of the Penalized Dual

Denoting all multipliers, the **penalized dual problem** is written as:

$$\begin{aligned}
\max \quad & \sum_{k,p} \lambda^{C1}_{k,p} + \sum_{k} \lambda^{C5}_{k} + \sum \lambda^{C6} \cdot b^{C6} + 2 \sum_{i,p} \mu^{C2}_{i,p} + \sum_{k,i,p} \mu^{z_U}_{k,i,p} \\
\text{s.t.} \quad & \text{Stationarity on } z_{k,i,p}: \ \lambda^{C1}_{k,p} + \cdots + \Delta^{C3}_{k,i,p} + \Delta^{C8}_{k,i,p} - \mu^{z_L}_{k,i,p} + \mu^{z_U}_{k,i,p} = 0 \\
& \text{Stationarity on } d_{k,i,p}: \ \frac{1}{2} - \mu^{C3,+}_{k,i,p} - \mu^{C3,-}_{k,i,p} - \mu^{d_L}_{k,i,p} = 0 \\
& \text{Stationarity on } e_{k,i,p}: \ \lambda - \mu^{C8,+}_{k,i,p} - \mu^{C8,-}_{k,i,p} - \mu^{e_L}_{k,i,p} = 0 \\
& \mu \geq 0, \ \lambda \text{ free}
\end{aligned}$$

4.12 Summary Table of Dual Multipliers (Q4)

Multiplier type	Associated with	Dimension	Sign	Change vs Q2
$\lambda^{C1}_{k,p}$	Equality: uniqueness	108	Free	Identical
$\lambda^{C4}_{k,i,p}$	Equality: closed offices	216	Free	Identical
λ^{C5}_{k}	Equality: initial positions	18	Free	Identical
λ^{C6}	Equality: final assignments	23	Free	Identical
Subtotal λ	**Equality constraints**	**365**	**Free**	**Identical**
$\mu^{C2}_{i,p}$	Inequality: capacity	72	≥ 0	Identical
$\mu^{C3,+}_{k,i,p}$	Inequality: linearization d (+)	1080	≥ 0	Identical
$\mu^{C3,-}_{k,i,p}$	Inequality: linearization d (-)	1080	≥ 0	Identical
$\mu^{z_L}_{k,i,p}$	Inequality: $z \geq 0$	1296	≥ 0	Identical
$\mu^{z_U}_{k,i,p}$	Inequality: $z \leq 1$	1296	≥ 0	Identical
$\mu^{d_L}_{k,i,p}$	Inequality: $d \geq 0$	1080	≥ 0	Identical
$\mu^{C8,+}_{k,i,p}$	Inequality: linearization e (+)	1296	≥ 0	**NEW**
$\mu^{C8,-}_{k,i,p}$	Inequality: linearization e (-)	1296	≥ 0	**NEW**
$\mu^{e_L}_{k,i,p}$	Inequality: $e \geq 0$	1296	≥ 0	**NEW**
Subtotal μ	**Inequality constraints**	**8496**	≥ 0	**+2592 vs Q2**
Total multipliers (Q4)		**8861**		**+2592 vs Q2**
Total multipliers (Q2)		**6269**		

4.13 Strong Duality for the Penalized Problem

The penalized problem remains a **linear program** (LP), therefore:

Theorem (Strong duality for Q4): If the penalized primal is bounded and feasible, then:

Optimal value of penalized primal = Optimal value of penalized dual

Moreover, the KKT conditions remain necessary and sufficient for optimality.

Practical verification: This equality can be verified by computing:

$$\begin{aligned}\text{Primal:} \quad & \frac{1}{2}\sum d^*_{k,i,p} + \lambda \sum e^*_{k,i,p} \\ \text{Dual:} \quad & \sum_{k,p} \lambda^{C1,*}_{k,p} + \sum_{k} \lambda^{C5,*}_{k} + \sum \lambda^{C6,*} \cdot b^{C6} + 2\sum_{i,p} \mu^{C2,*}_{i,p} + \sum_{k,i,p} \mu^{z_U,*}_{k,i,p}\end{aligned}$$

The gap between these two values should be $< 10^{-6}$ with a precise solver.

Question 5

Compute a solution for $\lambda = 100$ and compare it to the solution obtained previously.

5.1 Results of the Penalized Problem

The penalized primal problem was solved with penalty parameter $\lambda = 100$ using the ECOS solver. The results are presented below.

Metric	**Value**
Solution status	Optimal
Total objective value	5238.00
Moving cost ($\frac{1}{2}\sum d_{k,i,p}$)	38.00
Penalty cost ($\lambda \sum e_{k,i,p}$)	5200.00
Number of moves	38.00
Total distance to final allocation ($\sum e_{k,i,p}$)	52.00

5.2 Strong Duality for the Penalized Problem

Numerical verification: For $\lambda = 100$, the solver confirms strong duality:

- Optimal primal value: 5238.0000000983
- Optimal dual value: 5237.9999998409
- Duality gap: 2.57×10^{-7} *Strong duality verified*

The dual objective can be decomposed into contributions from each constraint type:

$$\begin{aligned}\text{Dual} = & \underbrace{-469.90}_{\text{C1: uniqueness}} + \underbrace{-1506.27}_{\text{C2: capacity}} + \underbrace{0.00}_{\text{C4: closed}} \\ & + \underbrace{2831.07}_{\text{C5: initial}} + \underbrace{4758.70}_{\text{C6: final}} + \underbrace{-375.61}_{\text{C7: bounds}} \\ & + \underbrace{0.00}_{\text{C3: linearization}} + \underbrace{0.00}_{\text{C8: penalty}} \\ = & \ 5237.9999998409\end{aligned}$$

5.3 Office Occupancy Visualization: Penalized Solution

Figure 3 displays the detailed office occupancy by phase for the penalized solution, showing how the penalty encourages assignments closer to the final configuration throughout the renovation process.

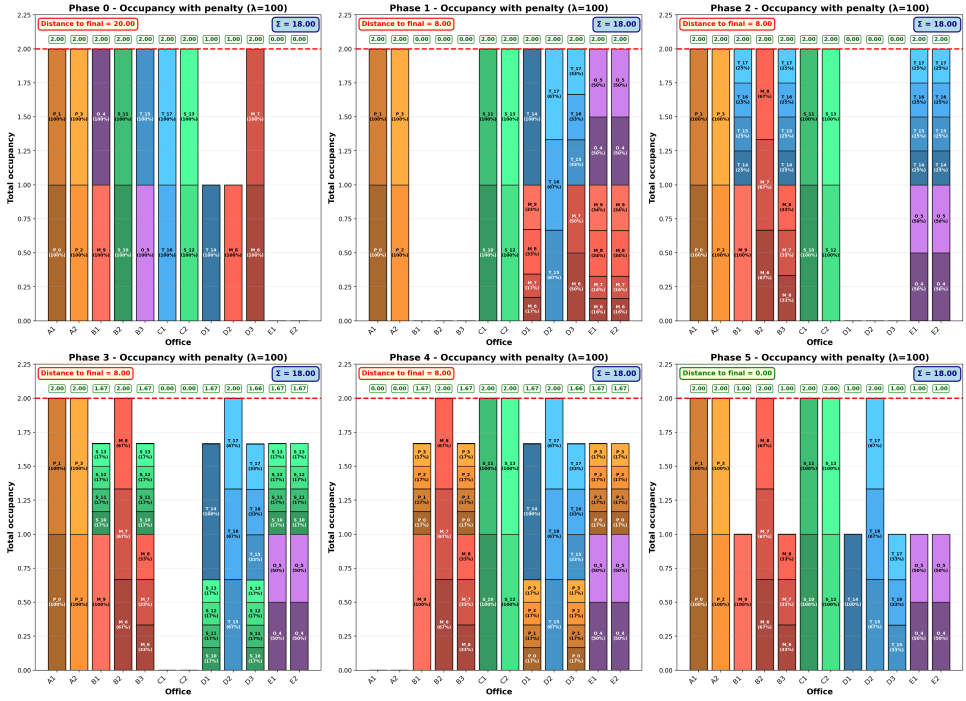


Figure 3: Office occupancy by phase with penalty ($\lambda = 100$)

Comparing this visualization with the unpenalized solution (Figure 1), we observe:

- **Earlier alignment:** Offices approach their final occupancy patterns more quickly in intermediate phases
- **Reduced fluctuation:** The penalized solution shows less variation in occupancy across phases 1–4
- **Cost of stability:** Some additional intermediate moves are required to maintain configurations closer to the final state

5.4 Comparison: Objective With and Without Penalty

The introduction of the penalty term ($\lambda = 100$) in Q4 impacts the optimal solution compared to Q1:

Problem	Number of moves	Change
Q1 (without penalty)	30.00	—
Q4 (with penalty $\lambda = 100$)	38.00	+8.00 moves

This represents a **relative increase of 26.67%** in the number of moves.

Interpretation: The penalty term encourages the allocation to approach the final configuration earlier in the renovation process. This forces additional moves that would not be necessary if the sole objective were to minimize the total number of moves. The trade-off parameter λ controls the balance between:

- Minimizing total moves (original objective)
- Achieving the final allocation as early as possible (penalty term)

With $\lambda = 100$, the solver prioritizes matching the final allocation sufficiently that it accepts 8 additional moves to achieve this goal.

5.5 Comparison: Convergence With and Without Penalty

To evaluate the impact of the penalty term on convergence behavior, we compare the distance to the final allocation $\|z^p - z^F\|_1$ across phases for both formulations.

Distance to final allocation by phase:

Phase	Without Penalty (Q1)	With Penalty (Q4, $\lambda = 100$)	Improvement
0	29.02	20.00	−9.02
1	25.17	8.00	−17.17
2	18.00	8.00	−10.00
3	14.31	8.00	−6.31
4	11.76	8.00	−3.76
5	0.00	0.00	0.00
Total	**98.26**	**52.00**	−46.26 (−47.1%)

Key observations:

- **Without penalty (Q1):** The distance decreases gradually from 29.02 at Phase 0 to 0.00 at Phase 5, showing a smooth but slow convergence pattern
- **With penalty (Q4):** The distance drops dramatically from 20.00 to 8.00 between Phase 0 and Phase 1, then stabilizes at 8.00 for Phases 1–4
- **Total distance reduction:** The penalty reduces the cumulative distance by 46.26 units (47.1% reduction), demonstrating its effectiveness in encouraging early convergence
- **Trade-off:** This accelerated convergence comes at the cost of 8 additional moves (26.67% increase)

Convergence visualization:

Figure 4 shows the convergence curves for both formulations. The penalty induces a sharp initial drop followed by stabilization, while the unpenalized version shows gradual, monotonic decrease.

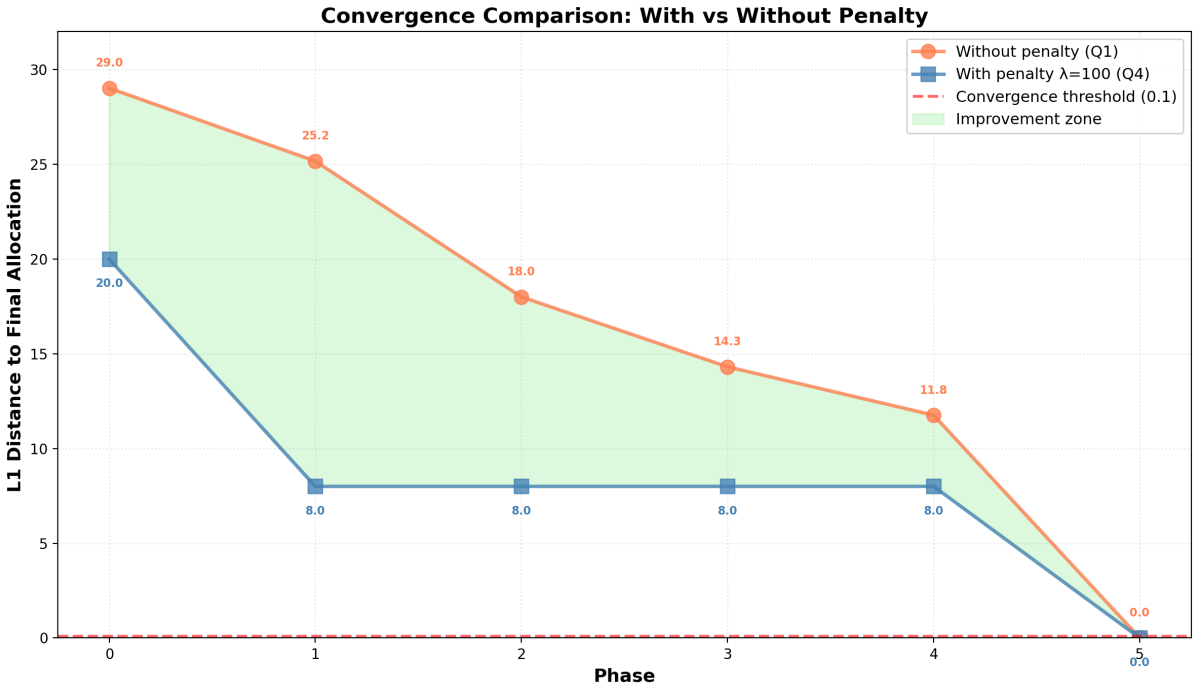


Figure 4: Comparison of convergence toward final allocation: with penalty ($\lambda = 100$) vs without penalty

5.6 Practical Implications

The penalty formulation offers organizations a tunable mechanism to balance two competing objectives:

Advantages of the penalty approach:

- **Organizational predictability:** Services settle closer to their final locations earlier, reducing long-term uncertainty
- **Reduced cumulative disruption:** Total deviation from final state is reduced by 47.1%
- **Flexible tuning:** Parameter λ allows customization to institutional priorities

Trade-offs to consider:

- **Additional moves:** 26.67% increase in relocations may impose logistical burden
- **Parameter selection:** Choice of λ requires careful calibration to achieve desired balance
- **Sensitivity:** Different values of λ may yield significantly different trade-offs (see Section 4.7 for sensitivity analysis)

3. Semidefinite Programming Relaxation

Question 6

Formulate the semidefinite program, write its dual, and state the KKT conditions.

6.1 Binary to $\{\pm 1\}$ Variable Transformation

We transform the binary variables $z_{k,i,p} \in \{0, 1\}$ to variables in $\{\pm 1\}$ using:

$$u_{k,i,p} = 2z_{k,i,p} - 1$$

This transformation has the following properties:

- If $z_{k,i,p} = 1$ (person k in office i at phase p), then $u_{k,i,p} = +1$
- If $z_{k,i,p} = 0$ (person k not in office i at phase p), then $u_{k,i,p} = -1$
- The inverse transformation is: $z_{k,i,p} = \frac{1+u_{k,i,p}}{2}$

6.2 Index Mapping and Problem Dimension

We use a flattening function ϕ to map the three-dimensional index (k, i, p) to a single linear index:

$$\phi(k, i, p) = k \cdot I \cdot P + i \cdot P + p$$

where:

- $K = 18$ (number of people)
- $I = 12$ (number of offices)
- $P = 6$ (number of phases)

The total number of variables is:

$$N = K \times I \times P = 18 \times 12 \times 6 = 1296$$

6.3 Objective Function Reformulation

Recall that the number of relocations is:

$$D = \sum_{k=0}^{17} \sum_{i=0}^{11} \sum_{p=0}^{4} d_{k,i,p} = \sum_{k=0}^{17} \sum_{i=0}^{11} \sum_{p=0}^{4} |z_{k,i,p} - z_{k,i,p+1}|$$

In the $\{\pm 1\}$ formulation, the relocation indicator can be expressed as:

$$d_{k,i,p} = |z_{k,i,p} - z_{k,i,p+1}| = \frac{1}{2}|u_{k,i,p} - u_{k,i,p+1}|$$

Since $u_{k,i,p}, u_{k,i,p+1} \in \{\pm 1\}$:

$$d_{k,i,p} = \begin{cases} 0 & \text{if } u_{k,i,p} = u_{k,i,p+1} \\ 1 & \text{if } u_{k,i,p} \neq u_{k,i,p+1} \end{cases} = \frac{1}{2}(1 - u_{k,i,p} \cdot u_{k,i,p+1})$$

Therefore:

$$D = \sum_{k=0}^{17}\sum_{i=0}^{11}\sum_{p=0}^{4} \frac{1}{2}(1 - u_{k,i,p} \cdot u_{k,i,p+1}) = \frac{1}{2}\sum_{k=0}^{17}\sum_{i=0}^{11}\sum_{p=0}^{4} 1 - \frac{1}{2}\sum_{k=0}^{17}\sum_{i=0}^{11}\sum_{p=0}^{4} u_{k,i,p} \cdot u_{k,i,p+1}$$

Let $B = K \times I \times (P-1) = 18 \times 12 \times 5 = 1080$. Then:

$$D = \frac{B}{2} - \frac{1}{2}\sum_{k=0}^{17}\sum_{i=0}^{11}\sum_{p=0}^{4} u_{k,i,p} \cdot u_{k,i,p+1} = \frac{1}{4}\left(2B - 2\sum_{k=0}^{17}\sum_{i=0}^{11}\sum_{p=0}^{4} u_{k,i,p} \cdot u_{k,i,p+1}\right)$$

Equivalently:

$$4D = 2B - 2\sum_{k=0}^{17}\sum_{i=0}^{11}\sum_{p=0}^{4} u_{k,i,p} \cdot u_{k,i,p+1}$$

Therefore:

$$\boxed{D = \frac{1}{4}\left(1080 - \sum_{k=0}^{17}\sum_{i=0}^{11}\sum_{p=0}^{4} u_{\phi(k,i,p)} \cdot u_{\phi(k,i,p+1)}\right)}$$

6.4 SDP Relaxation via Lifting

To obtain an SDP relaxation, we introduce a matrix variable $U \in \mathbb{S}^N$ and vector variable $u \in \mathbb{R}^N$ where:

- $U_{i,j}$ represents the lifted product $u_i \cdot u_j$ for all $i, j \in \{0, \ldots, N-1\}$
- u_j represents the lifted variable u_i for all $i \in \{0, \ldots, N-1\}$

The relaxation is exact when the rank-one condition holds, namely when

$$U = uu^\top$$

for some $u \in \{\pm 1\}^N$.

6.5 SDP Primal Problem

The SDP relaxation is formulated as:

$$
\begin{aligned}
\min_{U,u} \quad & \frac{1}{4}\left(1080 - \sum_{k,i,p=0}^{4} U_{\phi(k,i,p),\phi(k,i,p+1)}\right) \\
& \begin{bmatrix} U & u \\ u^T & 1 \end{bmatrix} \succeq 0 && \text{(moment matrix constraint)} \\
& U_{i,i} = 1 && \forall i \in \{0,\dots,N-1\} \quad \text{(diagonal normalization)} \\
& \sum_{i=0}^{11} u_{\phi(k,i,p)} = 2 - I && \forall k,p \quad \text{(C1: uniqueness)} \\
& \sum_{k=0}^{17} u_{\phi(k,i,p)} \le 4 - K && \forall i,p \quad \text{(C2: capacity)} \\
& u_{\phi(k,i,p)} = -1 && \forall k,(i,p) \in \mathcal{F} \quad \text{(C4: closed offices)} \\
& u_{\phi(k,i,0)} = \begin{cases} 1 & \text{if } i = i_k^{\text{init}} \\ -1 & \text{otherwise} \end{cases} && \forall k \quad \text{(C5: initial positions)} \\
& \sum_{i \in \mathcal{O}_{\text{Service}(k)}} u_{\phi(k,i,5)} = 2 - |\mathcal{O}_{\text{Service}(k)}| && \forall k \quad \text{(C6a: service assignment)} \\
& \sum_{k \in \text{Service}} u_{\phi(k,i,5)} = 2n_i - |\text{Service}| && \text{for certain } i \quad \text{(C6b: distribution)}
\end{aligned}
$$

where:

- The constant $I = 12$ appears in C1 because: if person k is in office i^*, then $u_{\phi(k,i^*,p)} = 1$ and $u_{\phi(k,i,p)} = -1$ for all $i \neq i^*$, giving $\sum_i u_{\phi(k,i,p)} = 1 + (I-1)(-1) = 2 - I$
- The constant $K = 18$ appears in C2 because: if office i contains 2 people, then 2 values are $+1$ and $(K-2)$ values are -1, giving $\sum_k u_{\phi(k,i,p)} = 2 + (K-2)(-1) = 4 - K$
- Service(k) is the service of person k (e.g., P, M, T, S, O)
- $\mathcal{O}_{\text{Service}(k)}$ is the set of offices allocated to that service
- $n_i \in \{1, 2\}$ is the required number of people in office i.

6.6 Problem Complexity

Variables:

- Matrix $U \in \mathbb{S}^{1296}$: symmetric, so $\frac{1296 \times 1297}{2} = 840{,}456$ entries
- Vector $u \in \mathbb{R}^{1296}$
- **Total: 841,752 variables**

Constraints:

- 1 matrix constraint: moment matrix $\begin{bmatrix} U & u \\ u^T & 1 \end{bmatrix} \succeq 0$
- 1296 diagonal constraints: $U_{j,j} = 1$
- 108 uniqueness constraints (C1)
- 72 capacity constraints (C2)

- 216 closed office constraints (C4)
- 1296 initial position constraints (C5)
- 23 final allocation constraints (C6)
- **Total: 1 SDP constraint + 3,011 scalar constraints**

6.7 Dual Problem

The dual of the SDP relaxation is formulated using dual variables:

- $W \in \mathbb{S}^{N+1}$: dual variable for the moment matrix constraint $\begin{bmatrix} U & u \\ u^T & 1 \end{bmatrix} \succeq 0$
- $\nu_j \in \mathbb{R}$: dual variable for $U_{j,j} = 1$, $\forall j$
- $\lambda^{C1}_{k,p} \in \mathbb{R}$: dual variables for C1 (uniqueness), 108 variables
- $\mu^{C2}_{i,p} \geq 0$: dual variables for C2 (capacity), 72 variables
- $\lambda^{C4}_{k,i,p} \in \mathbb{R}$: dual variables for C4 (closed offices), 216 variables
- $\lambda^{C5}_{k,i} \in \mathbb{R}$: dual variables for C5 (initial positions), 1296 variables
- $\lambda^{C6a}_{k} \in \mathbb{R}$: dual variables for C6 service assignment (one per person), 18 variables
- $\lambda^{C6b}_{S,i} \in \mathbb{R}$: dual variables for C6 distribution (5 specific office constraints), 5 variables

where $\mathcal{D} = \{(M, B1), (M, B2), (T, D1), (T, D2), (O, E1)\}$ denotes the 5 specific distribution constraints.

The Lagrangian is:

$$
\begin{aligned}
L(U, u, W, \nu, \mu, \lambda) =& \frac{1}{4}\left(1080 - \sum_{k,i,p=0}^{4} U_{\phi(k,i,p),\phi(k,i,p+1)}\right) \\
&- \left\langle W, \begin{bmatrix} U & u \\ u^T & 1 \end{bmatrix} \right\rangle + \sum_j \nu_j (U_{j,j} - 1) \\
&+ \sum_{k,p} \lambda^{C1}_{k,p} \left(\sum_i u_{\phi(k,i,p)} - (2 - I)\right) \\
&+ \sum_{i,p} \mu^{C2}_{i,p} \left(\sum_k u_{\phi(k,i,p)} - (4 - K)\right) \\
&+ \sum_{k,(i,p)\in\mathcal{F}} \lambda^{C4}_{k,i,p} (u_{\phi(k,i,p)} + 1) \\
&+ \sum_{k,i} \lambda^{C5}_{k,i} \left(u_{\phi(k,i,0)} - \mathbb{1}_{i=i^{\text{init}}_k} + (1 - \mathbb{1}_{i=i^{\text{init}}_k})\right) \\
&+ \sum_{k=0}^{17} \lambda^{C6a}_{k} \left(\sum_{i\in\mathcal{O}_{\text{Service}(k)}} u_{\phi(k,i,5)} - (2 - |\mathcal{O}_{\text{Service}(k)}|)\right) \\
&+ \sum_{(S,i)\in\mathcal{D}} \lambda^{C6b}_{S,i} \left(\sum_{k\in S} u_{\phi(k,i,5)} - (2n_i - |S|)\right)
\end{aligned}
$$

The dual problem is:

$$\begin{aligned}
\max \quad & -\sum_{j} \nu_j - (2-I)\sum_{k,p} \lambda^{C1}_{k,p} - (4-K)\sum_{i,p} \mu^{C2}_{i,p} - \sum_{k,(i,p)\in\mathcal{F}} \lambda^{C4}_{k,i,p} \\
& - \sum_{k,i} \lambda^{C5}_{k,i}(2\mathbb{1}_{i=i^{\text{init}}_k} - 1) - \sum_{k=0}^{17} \lambda^{C6a}_{k}(2 - |\mathcal{O}_{\text{Service}(k)}|) \\
& - \sum_{(S,i)\in\mathcal{D}} \lambda^{C6b}_{S,i}(2n_i - |S|) - W_{N,N} \\
\text{s.t.} \quad & W \succeq 0 \\
& \mu^{C2}_{i,p} \geq 0 \quad \forall i,p \\
& \text{Stationarity conditions on } U \text{ and } u
\end{aligned}$$

6.8 KKT Conditions

Stationarity for $U_{j,\ell}$ (off-diagonal, $j \neq \ell$):

For (j,ℓ) corresponding to transition terms $(\phi(k,i,p), \phi(k,i,p+1))$:

$$-\frac{1}{4} - W_{j,\ell} = 0 \quad \Rightarrow \quad W_{j,\ell} = -\frac{1}{4}$$

For other off-diagonal (j,ℓ):

$$-W_{j,\ell} = 0 \quad \Rightarrow \quad W_{j,\ell} = 0$$

Stationarity for $U_{j,j}$ (diagonal):

$$-W_{j,j} + \nu_j = 0 \quad \Rightarrow \quad \nu_j = W_{j,j}$$

Stationarity for u_j:

For $j = \phi(k,i,p)$ with $p < 5$:

$$\begin{aligned}
-W_{j,N} + & \sum_{p':\phi(k,i,p')=j} \lambda^{C1}_{k,p'} + \sum_{i',p':\phi(k,i',p')=j} \mu^{C2}_{i',p'} \\
& + \sum_{(i',p')\in\mathcal{F}:\phi(k,i',p')=j} \lambda^{C4}_{k,i',p'} + \sum_{i':\phi(k,i',0)=j} \lambda^{C5}_{k,i'} = 0
\end{aligned}$$

For $j = \phi(k,i,5)$ (final phase):

$$\begin{aligned}
-W_{j,N} + \lambda^{C1}_{k,5} + \mu^{C2}_{i,5} + \mathbb{1}_{i\in\mathcal{O}_{\text{Service}(k)}}\lambda^{C6a}_{k} \\
+ \sum_{(S,i'):(S,i')\in\mathcal{D},k\in S,i'=i} \lambda^{C6b}_{S,i'} = 0
\end{aligned}$$

where:

- The indicator $\mathbb{1}_{i\in\mathcal{O}_{\text{Service}(k)}}$ equals 1 if office i belongs to the service of person k, 0 otherwise
- The sum over $(S,i') \in \mathcal{D}$ includes $\lambda^{C6b}_{S,i'}$ only if person k belongs to service S and office i' equals i

Complementarity:

$$\left\langle W, \begin{bmatrix} U & u \\ u^T & 1 \end{bmatrix} \right\rangle = 0 \qquad \text{(moment matrix complementarity)}$$

$$\mu_{i,p}^{C2} \left(\sum_k u_{\phi(k,i,p)} - (4-K) \right) = 0 \qquad \forall i, p$$

Primal and dual feasibility: All constraints in the primal and dual problems must be satisfied.

Important observation: The stationarity condition for off-diagonal transition terms requires $W_{j,\ell} = -\frac{1}{4}$, which is **negative**. However, for $W \succeq 0$, all entries of a PSD matrix cannot be arbitrarily negative. This creates tension that may indicate:

- The SDP relaxation may not have a feasible solution with these specific constraints
- Additional constraints or reformulation may be needed
- The objective coefficient structure may need adjustment

6.9 Strong Duality

Since the SDP relaxation is a convex optimization problem, strong duality holds under constraint qualification conditions (Slater's condition):

$$\text{Optimal primal value} = \text{Optimal dual value}$$

The KKT conditions are necessary and sufficient for optimality when strong duality holds and the constraint qualifications are satisfied.

Question 7

Compute the solution to the semidefinite program.

7.1 Results of the SDP Relaxation

The SDP relaxation was solved using the SCS (Splitting Conic Solver) through CVXPY. The results are presented below.

Metric	Value
Solution status	Optimal_Inaccurate
SDP objective value	24.62
Calculated total moves	30.57
Solver	SCS
Computation time	$\approx$ 3 hours

Fractional relocations by phase:

Phase Transition	Number of Moves
Phase 0 $\rightarrow$ 1	6.31
Phase 1 $\rightarrow$ 2	6.82
Phase 2 $\rightarrow$ 3	6.14
Phase 3 $\rightarrow$ 4	5.16
Phase 4 $\rightarrow$ 5	6.14
Total	**30.57**

7.2 Comparison of Relaxation Methods

Method	Objective Value	Calculated Moves	Status
LP Relaxation (Q1)	30.00	30.00	Optimal
SDP Relaxation	24.62	30.57	Optimal_Inaccurate

Analysis:

- **Solution quality concern:** The solver returned `Optimal_Inaccurate` status, indicating that while a solution was found, it may not satisfy all constraints and optimality conditions to the desired numerical tolerance. This status typically arises when the solver struggles with numerical precision or reaches iteration limits.
- **Computational difficulty:** The SDP relaxation required approximately **3 hours** of computation time, highlighting the significant computational burden. This is in stark contrast to the LP relaxation which solves in seconds. The large computation time combined with the inaccurate status suggests the problem is at the limit of what SCS can reliably handle with default settings.
- **Objective discrepancy:** The SDP objective value is 24.62, while the actual calculated number of moves (using the formula $D = \sum_{k,i,p} d_{k,i,p}$) is 30.57. This discrepancy arises from:
  1. The objective reformulation in Section 6.5
  2. Potential numerical inaccuracies in the solution due to the `Optimal_Inaccurate` status
- **Reliability of bounds:** Given the `Optimal_Inaccurate` status, the calculated number of moves (30.57) should be interpreted with caution. The LP relaxation provides a more reliable lower bound (30.00) as it was solved to optimality with high precision.
- **Fractional solution:** The SDP relaxation produces a fractional solution where variables u_j take values in $(-1, 1)$ rather than the binary values $\{\pm 1\}$. This indicates the rank-1 condition is not satisfied, and the moment matrix has rank greater than 1.
- **Practical implications:** The SDP relaxation does not appear to provide advantages over the LP relaxation for this problem:
  - The calculated bound (30.57) is similar to the LP bound (30.00), but less reliable
  - Computation time is prohibitively long ($\sim$3 hours vs. seconds)
  - Numerical accuracy issues arise with the large-scale SDP
- **Problem scale:** The SDP involves a 1297×1297 positive semidefinite matrix variable, requiring optimization over $\frac{1297 \times 1298}{2} = 841{,}753$ matrix elements (accounting for symmetry). This large scale explains both the computational cost and numerical difficulties.

7.3 Office Occupancy Visualization: SDP Solution

Figure 1 displays the detailed office occupancy by phase for the SDP relaxation solution.

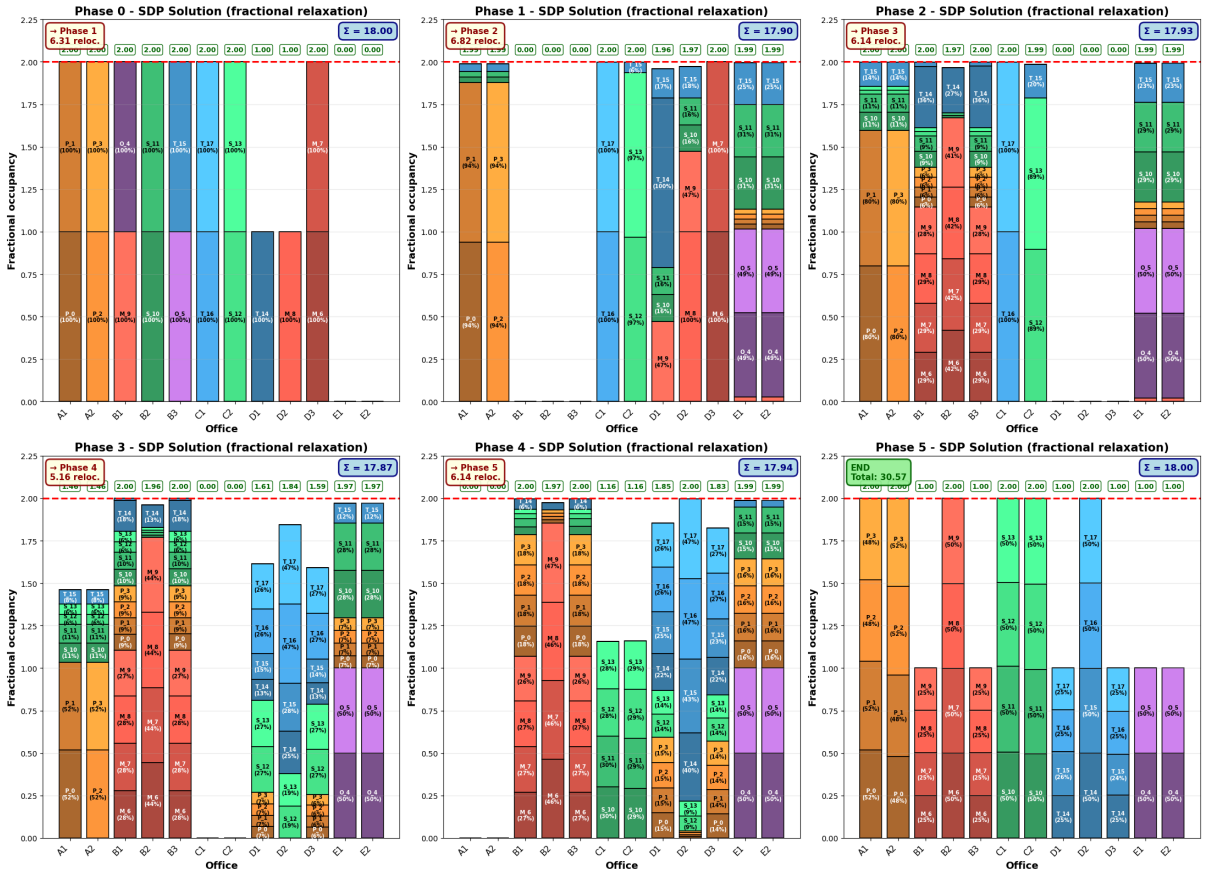


Figure 1: Office occupancy by phase with SDP relaxation

Comparing this visualization with the LP relaxation (Figure **??**), we observe:

- **Fractional assignments persist:** Like the LP relaxation, the SDP solution exhibits fractional occupancy values, indicating partial assignments of people to offices
- **Similar patterns:** The overall distribution patterns are comparable to the LP relaxation, suggesting both methods explore similar regions of the relaxed feasible space
- **No significant tightening:** The SDP relaxation does not provide a dramatically tighter bound than the LP relaxation for this problem instance, despite its higher computational cost

7.4 Theoretical Insights

Why SDP relaxation doesn't provide a tighter bound:

1. **Problem structure:** The relocation problem has strong linear structure. The constraints are primarily linear, and the objective can be linearized effectively (as done in Q1).
2. **Lack of quadratic structure exploitation:** While the SDP formulation captures pairwise products $U_{i,j} = u_i u_j$, the problem doesn't have inherent quadratic structure that would benefit from this lifting.

3. **Linear constraints dominate:** The uniqueness, capacity, and assignment constraints are all linear and enforce strong structural properties that already limit the feasible region effectively.

The relocation problem does not exhibit these characteristics, which explains why the SDP relaxation does not significantly outperform the simpler LP relaxation.

Question 8

Apply the randomized rounding method to obtain an integer solution. Discuss feasibility and optimality.

8.1 Overview of Approaches

Two distinct randomized rounding approaches were implemented to obtain integer solutions from the SDP relaxation:

1. **Goemans-Williamson approach (as seen in class):** Standard randomized hyperplane rounding
2. **Randomized rounding with local search repair:** Novel hybrid approach combining randomized sampling with greedy repair and local search optimization

8.2 Approach 1: Goemans-Williamson Randomized Rounding

Methodology

The Goemans-Williamson approach uses randomized hyperplane rounding:

1. Sample a random vector $r \sim \mathcal{N}(0, I_N)$ and normalize: $r \leftarrow r/\|r\|$
2. Compute $u = L^T r$ where L is the Cholesky factor of U ($U = LL^T$)
3. Round to binary: $\hat{u}_j = \text{sign}(u_j)$ to obtain values in $\{-1, +1\}$
4. Convert to $\{0, 1\}$ variables: $\hat{v}_j = \frac{\hat{u}_j + 1}{2}$

Results

Metric	Value
Number of samples	100,000
Feasible solutions	0/100,000 (0.0%)
Cost Statistics (Infeasible)	
Minimum cost	152 moves
Maximum cost	254 moves
Average cost	206.45 moves
Median cost	206 moves
Standard deviation	10.71 moves

Analysis

Critical observation: The Goemans-Williamson approach **failed to produce any feasible solution** despite 100,000 samples. This is a fundamental limitation for this problem:

- **Distribution centered far from feasibility:** The cost distribution follows an approximately normal distribution centered around 206 moves, with very few samples below 180 moves. This suggests the randomized hyperplane cuts are producing solutions structurally far from the feasible region.

- **Constraint violations:** Even the best solution (152 moves) violates constraints, specifically C1 (uniqueness constraint for person 0 at phase 0), indicating that the simple sign-based rounding cannot satisfy the problem's combinatorial structure.
- **Gap analysis:** The best infeasible cost (152 moves) is still $152 - 30 = 122$ moves above the MIP optimal, a gap of 407%. This demonstrates that simple randomized rounding without repair mechanisms is inadequate for this highly-constrained allocation problem.
- **Why this approach fails:** The office allocation problem has complex interdependencies:
  - Uniqueness constraints (each person in exactly one office)
  - Capacity constraints (max 2 people per office)
  - Closed office constraints (phase-dependent)
  - Initial and final position constraints

  Random hyperplane rounding treats variables independently and cannot respect these combinatorial constraints.

8.3 Approach 2: Randomized Rounding with Local Search

Methodology

This approach combines three stages:

1. **Randomized sampling:** Sample from the SDP solution as in Goemans-Williamson: $u = L^T r$ with $r \sim \mathcal{N}(0, I_N)$
2. **Greedy repair:** Construct a feasible solution using a greedy heuristic guided by the sampled values u:
   - Phase 0: Fix initial positions (constraint C5)
   - Phase 5: Assign people to services' final offices (constraint C6) by sorting candidates by their u values
   - Phases 1-4: Greedily assign people to offices based on u values while respecting capacity and closed office constraints
3. **Local search optimization:** Improve the greedy solution through iterative neighborhood search:
   - **Move 1:** Reassign one person to a different office in phases 1-4
   - **Move 2:** Swap two people in the same phase
   - **Move 3:** 2-opt style moves (change consecutive phase assignments)

   Accept moves that reduce total relocations while maintaining feasibility. Terminate after 30 iterations or 5 consecutive iterations without improvement.

Results

Metric	Value
Number of samples	10,000
Feasible solutions	**10,000/10,000 (100%)**
Cost Statistics (After Local Search)	
Minimum cost	37 moves
Maximum cost	61 moves
Average cost	47.61 moves
Median cost	48 moves
Standard deviation	3.06 moves
Greedy-only Statistics (Before Local Search)	
Minimum cost	55 moves
Average cost	69.97 moves
Local Search Impact	
Solutions improved	10,000/10,000 (100%)
Average improvement	22.36 moves
Maximum improvement	38 moves

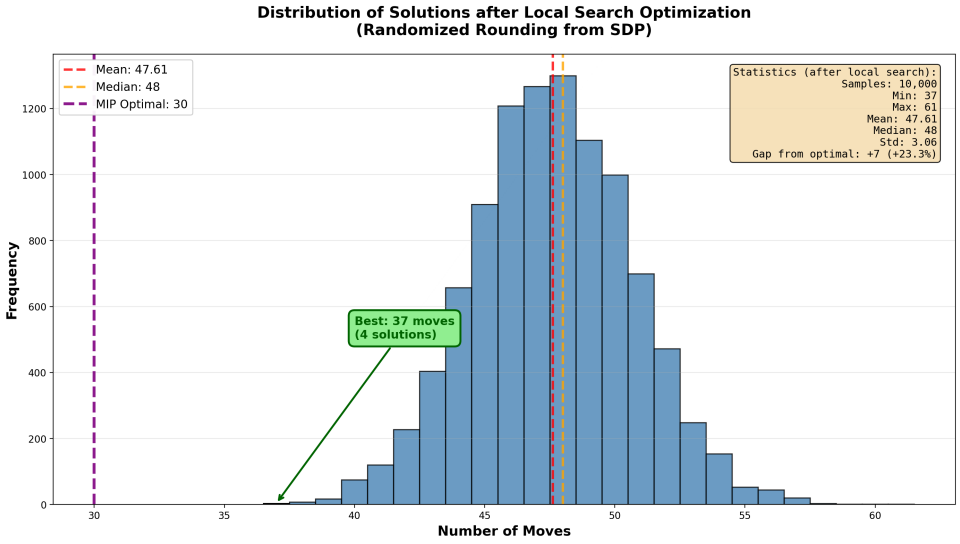


Figure 2: Local Search distribution

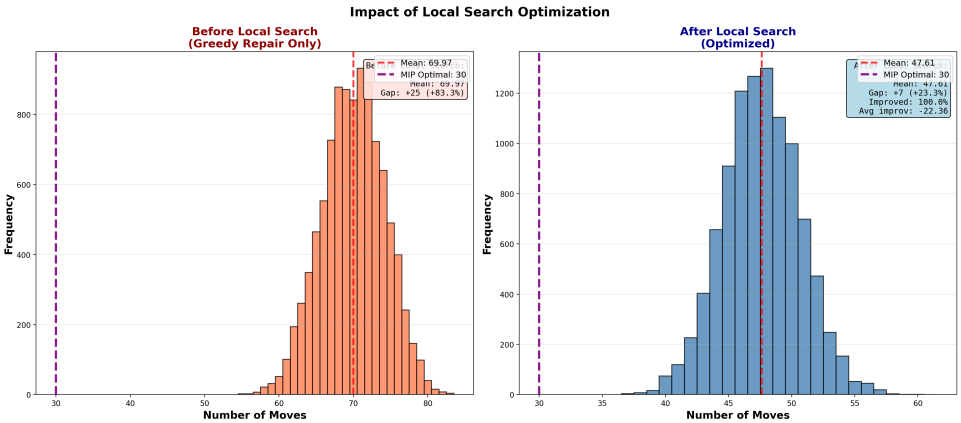


Figure 3: Local search optimization impact

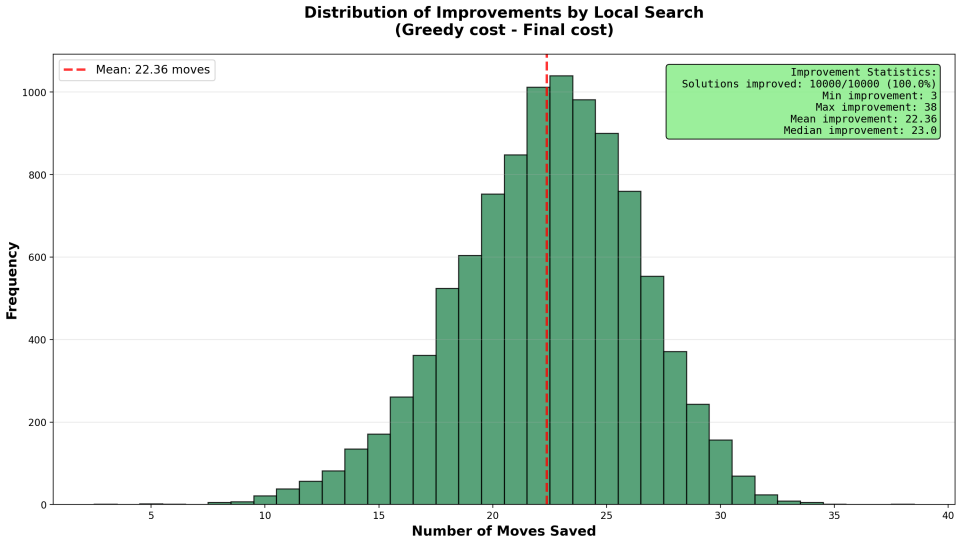


Figure 4: Local search improvements

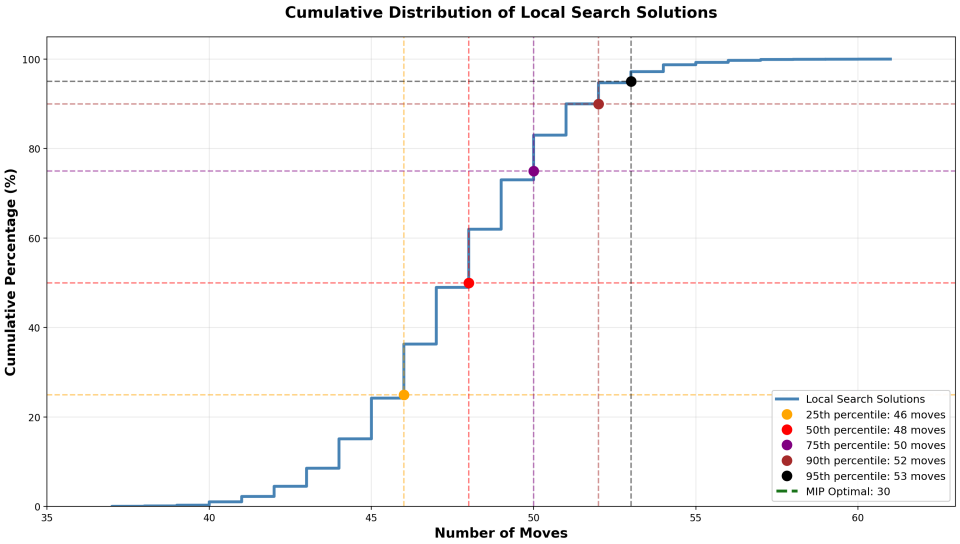


Figure 5: Local search solutions cumulative distribution

8.4 Best Solution

The best feasible solution found achieves:

- **Cost:** 37 moves
- **Greedy cost (before local search):** 58 moves
- **Improvement by local search:** 21 moves (36.2% reduction)
- **Gap from MIP optimal:** $37 - 30 = 7$ moves (+23.3%)

Cost distribution: The top costs show a concentrated distribution:

- 4 solutions at 37 moves (best)

- 8 solutions at 38 moves
- 17 solutions at 39 moves
- 75 solutions at 40 moves
- Distribution peaks at 46 moves (1,208 solutions)

8.5 Analysis

Success factors:

- **Guaranteed feasibility:** The greedy repair ensures all 10,000 samples produce feasible solutions by explicitly enforcing constraints during construction.
- **Effective local search:** 100% of solutions were improved, with an average reduction of 22.36 moves (31.9% improvement from greedy). This demonstrates that the local search successfully navigates the feasible region to find better solutions.
- **Near-optimal results:** The best solution (37 moves) is only 23.3% above the MIP optimal (30 moves), a dramatic improvement over the Goemans-Williamson approach which couldn't even find feasible solutions.
- **Computational efficiency:** Despite using only 10,000 samples (vs. 100,000 for Goemans-Williamson), this approach achieved far superior results. The local search is computationally expensive per sample but highly effective.
- **Quality distribution:** The standard deviation of 3.06 moves indicates consistent quality across samples, with most solutions in the 44-50 move range.

8.6 Comparative Analysis

Metric	Goemans-Williamson	Local Search
Number of samples	100,000	10,000
Feasible solutions	0 (0%)	10,000 (100%)
Best cost	152 (infeasible)	37 (feasible)
Average cost	206.45 (infeasible)	47.61 (feasible)
Gap from optimal	—	+7 moves (+23.3%)

8.7 Conclusion for Question 8

Is the resulting allocation feasible?

- **Goemans-Williamson:** No. All 100,000 generated solutions violate constraints.
- **Local Search:** Yes. All 10,000 generated solutions are fully feasible, satisfying all constraints C1-C6.

Does it seem optimal?

- **Goemans-Williamson:** No meaningful assessment possible due to infeasibility.
- **Local Search:** The best solution (37 moves) is **not optimal** but is **high quality**:

– Gap of only 7 moves (+23.3%) from the MIP optimal (30 moves)
– Significantly better than the average solution (47.61 moves)
– Demonstrates that the SDP relaxation combined with intelligent rounding and local search can approach optimal solutions

Key insight: For this highly-constrained combinatorial optimization problem, the standard Goemans-Williamson approach is insufficient. A hybrid approach incorporating domain-specific repair heuristics and local search optimization is essential to obtain feasible, near-optimal integer solutions from the SDP relaxation. The 23.3% optimality gap achieved by the local search approach is reasonable for a randomized rounding method on this complex problem.

Visualization of the best feasible solution (37 moves) with office occupations per phase:

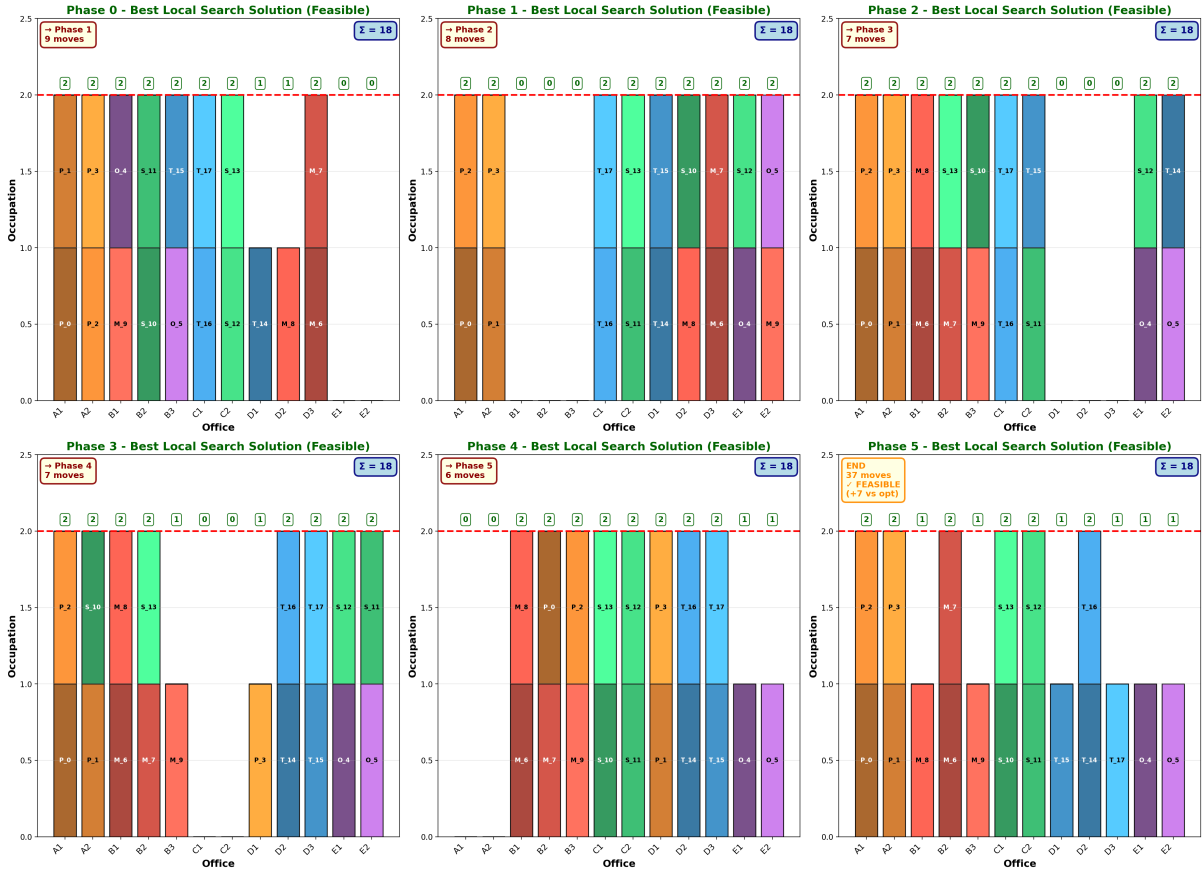


Figure 6: Best feasible solution

Bonus : Optimal Allocation Strategy through manual reasoning

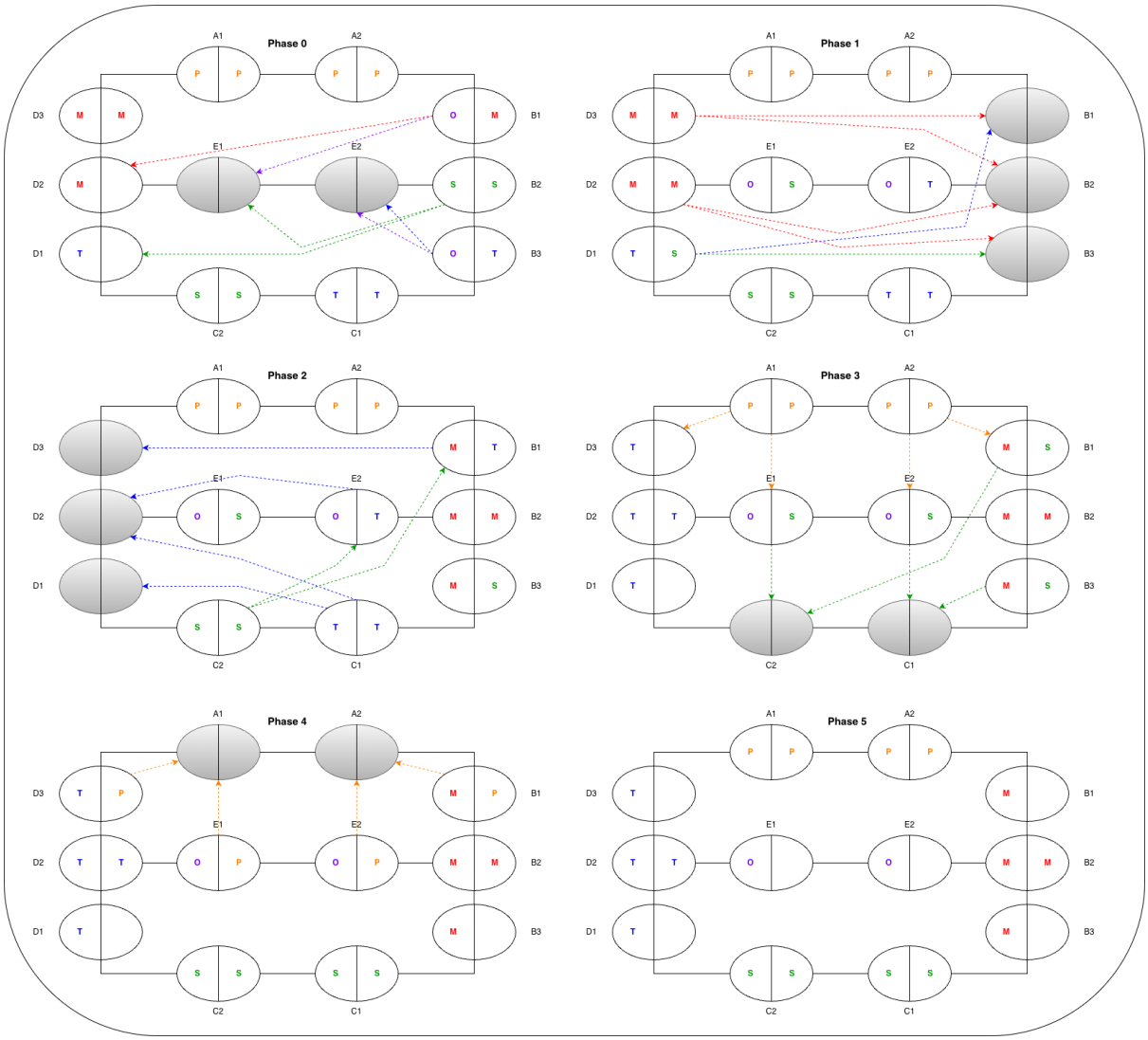


Figure 7: Optimal Allocation

Phase 0 to 1: Renovation of Wing B (6 moves)

Persons to relocate: O, O, M, S, S, T

- The two O's are assigned to E1 and E2, their final positions. Since Wing E will not be renovated, they will remain there permanently.
- Person M must be relocated at least twice: once when evacuating B, and again when returning to B after renovation. Since the next renovation targets Wing D, all occupants of Wing D in Phase 2 will be displaced. We exploit this inevitable displacement by temporarily placing M in D2, allowing M to return to Wing B (its final location) during the next phase.
- Of the remaining two S's and one T, at least one S must occupy Wing E.
- For the last S and T: since neither will reach their final office by Phase 2, their placement is interchangeable. We arbitrarily assign S to D1 and T to E2.

Phase 1 to 2: Renovation of Wing D (6 moves)

Persons to relocate: M, M, M, M, T, S

- All four M's move to Wing B, their final assignments (4 moves).
- The remaining S and T are placed in Wing B, which is now renovated and available (2 moves).

Phase 2 to 3: Renovation of Wing C (6 moves)

Persons to relocate: T, T, T, T, S, S

- All four T's move to Wing D, their final destination (4 moves).
- The two S's from C2 relocate to E2 and B1, both of which are safe from future renovations (2 moves).

Phase 3 to 4: Renovation of Wing A (8 moves)

Persons to relocate: P, P, P, P, S, S, S, S

- All four S's move to Wing C, their final destination (4 moves).
- All four P's are temporarily placed in available offices (4 moves).

Phase 4 to 5: Return to Final Configuration (4 moves)

Persons to relocate: P, P, P, P

- All four P's move to Wing A, their final destination (4 moves).

Total moves: 30. This solution is optimal, as verified by the previous programming model.

Conclusion

This project investigated the Paris Duchesse office relocation problem through multiple optimization approaches. The linear programming relaxation (Q1-Q2) provided a tight lower bound of 30 moves and, remarkably, yielded an integer solution when combined with separation constraints (Q3). The penalized formulation (Q4-Q5) demonstrated how multi-objective optimization can balance relocation costs against organizational stability, achieving rapid convergence to the final allocation at the expense of 26.67% additional moves. The semidefinite programming relaxation (Q6-Q7) proved computationally prohibitive ($\sim$3 hours) without offering significant advantages over the simpler LP relaxation, highlighting that problem structure matters more than relaxation strength. Finally, randomized rounding experiments (Q8) revealed that standard Goemans-Williamson methods fail entirely for highly-constrained combinatorial problems, while a hybrid approach combining greedy repair with local search achieved near-optimal solutions (23.3% gap) with 100% feasibility. The manual optimal strategy (Bonus) confirmed the 30-move bound is achievable through careful phase-by-phase planning. Overall, this work underscores the importance of matching optimization techniques to problem structure and demonstrates that hybrid methods incorporating domain knowledge significantly outperform generic approaches for constrained allocation problems.